

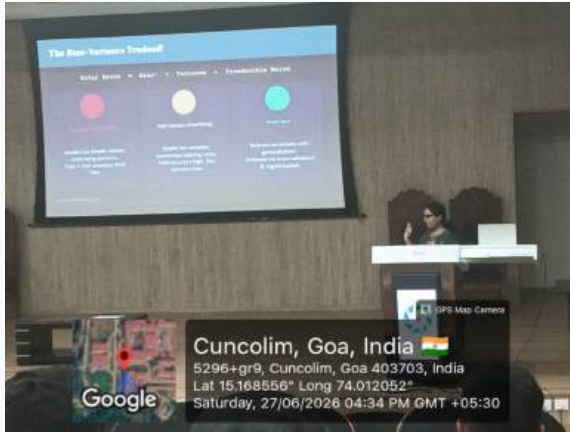


Dr. Vishwanath Karad
MIT WORLD PEACE
UNIVERSITY | PUNE
TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

Dr. Vishwanath Karad MIT - World Peace University
Faculty of Engineering and Technology
School of Computer Science and Engineering
Department of Computer Science and Applications
National Academic Immersion Program (NAIP)
National Institute of Technology, Goa
26 June To 7 July 2026

Submitted by :- Vedant Dilip Khedkar
PRN :- 1272240644
TYBCA-B

Day- Saturday	Date- 27th June 2026
Session 1	
Topic	Introduction to AI/ML/DL
Guest Name	Mr. Devendra Kuvelkar
Time	11:30-13:00
	



Topic Covered

An introductory overview of Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL). It explored foundational AI characteristics and categories, alongside ML paradigms like Supervised, Unsupervised, and Reinforcement learning, with specific emphasis on Linear and Logistic Regression.

Summary

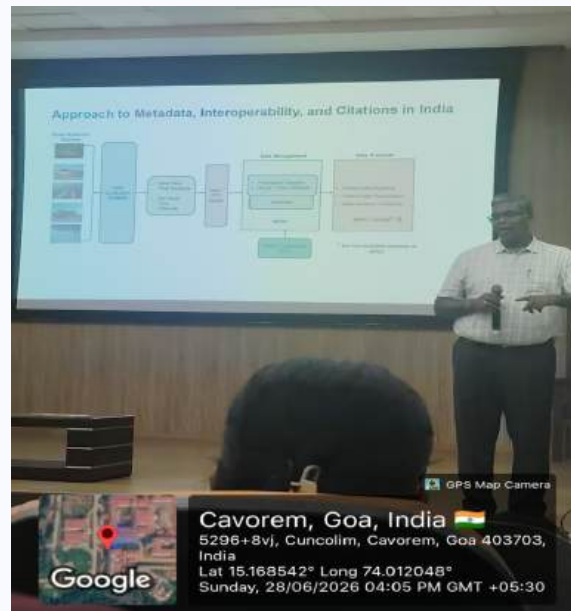
This session established a solid groundwork by detailing the core concepts of AI, ML, and DL. It mapped the relationships between them: AI as the overarching field, ML as the data-learning subset, and DL as the neural network-driven branch for complex problem-solving. The lecture clarified various AI classifications, distinguishing task-specific systems from broader intelligent machines. Participants also explored Supervised, Unsupervised, and Reinforcement Learning paradigms through practical, real-world examples to enhance comprehension. A major portion of the class was dedicated to implementing Linear and Logistic Regression models. Participants covered their underlying principles, use cases, and the differences between classifying data and predicting continuous values. Through hands-on

	exercises covering data preparation, training, evaluation, and result interpretation, attendees developed foundational machine learning skills.
Session 2	
Topic	Fundamentals of Machine Learning and Deep Learning
Guest Name	Ms. Siddhi Naik
Time	14:00-15:30
Topic Covered	Supervised Algorithms discussed included Linear/Logistic Regression, Decision Trees, Random Forest, SVM, KNN, and Naive Bayes. Unsupervised Algorithms encompassed K-Means, DBSCAN, PCA, Autoencoders, and Apriori. Deep Learning topics covered its definition, its relationship with AI/ML, and practical applications.
Summary	1. Supervised Learning: These algorithms train on labeled data. Decision Trees act as flowcharts, while Random Forests aggregate multiple trees to prevent overfitting. SVMs identify the optimal boundary line between complex data categories, and KNN classifies data based on its closest neighbors. Naive Bayes is a probability-based algorithm that assumes feature independence, making it highly effective for text sorting like spam filters. 2. Unsupervised Learning: These models uncover hidden patterns in raw, unlabeled data. K-Means assigns data to a specific number of clusters, whereas DBSCAN dynamically identifies dense clusters of any shape and ignores noise. PCA compresses

	massive datasets by reducing variables while preserving core information. Autoencoders compress and reconstruct data, which is useful for anomaly detection, and the Apriori algorithm finds hidden connections like frequently grouped purchases. 3. Deep Learning: The AI ecosystem operates like nesting dolls, with AI encompassing ML, which in turn encompasses DL. DL uses multi-layered neural networks inspired by the human brain, excelling at processing unstructured data like text, audio, and images for modern tech like self-driving systems and real-time translation.
Session 3	
Topic	Model Optimization: Methodologies and Best Practices
Guest Name	Mr. Devendra Kuvelkar
Time	16:00-17:30
Topic Covered	1. Deep Dive into Supervised Learning: Utilizing discriminative features and managing the bias-variance trade-off. 2. Diagnosing Model Fit: Differentiating underfitting and overfitting. 3. Strategy and Evaluation: Selecting algorithms and model evaluation best practices.
Summary	1. Deep Dive into Supervised Learning: Optimization requires moving beyond basic predictions to understand the learning

	process. This involves separating signal from noise by focusing on discriminative features that distinctly separate data classes. Practitioners must also balance the bias-variance trade-off—minimizing both the broad, inaccurate assumptions of bias and the hyper-sensitive memorization of variance to reduce total error. 2. Diagnosing Model Fit: Similar to a "Goldilocks problem," underfitting (high bias) means the model is too simple and performs poorly across all data. Overfitting (high variance) means the model perfectly memorizes training data but fails completely on new, unseen data. 3. Strategy and Evaluation: Real-world application requires a strategic approach to algorithm selection, which depends heavily on dataset size, problem complexity, and the need for interpretability. For instance, a highly transparent Decision Tree might be chosen over a complex Deep Neural Network if predictions must be clearly explained to stakeholders.
Session 4	
Topic	<i>(No Session 4 Scheduled)</i>
Guest Name	<i>(N/A)</i>
Time	<i>(N/A)</i>
Topic Covered	<i>(N/A)</i>
Summary	<i>(N/A)</i>

Day- Sunday	Date- 28th June 2026
Session 1	
Topic	<i>Nature-Inspired Optimization: From Evolution to Swarm Intelligence</i>
Guest Name	<i>Dr. Damodar Reddy</i>
Time	<i>09:30-11:00</i>



Topic Covered

1. Beyond Standard Search: Evolutionary Algorithms and Simulated Annealing.

2. The Power of Swarm Intelligence: Collective behaviors, system characteristics, and search mechanics.

3. Deep Dive: Particle Swarm Optimization (PSO), personal/global bests, and velocity updates.

4. Real-World Applications: Practical uses

	of swarm optimization.
Summary	1. Beyond Standard Search: When traditional mathematics struggles with massive problems, strategies inspired by biology and physics are utilized. Genetic Algorithms apply Darwinian evolution, mutating and breeding a population of solutions until the fittest one emerges. Simulated Annealing mimics metal cooling; it begins with wild, random data jumps to explore widely, then slowly "cools" to fine-tune and lock in on the best answer. 2. Swarm Intelligence: This concept studies how collective behaviors in nature, like ant colonies, solve complex logistics without central leadership. These systems use simple agents following basic rules to create highly adaptable networks that balance exploring new data areas with exploiting known optimal areas. 3. Particle Swarm Optimization (PSO): In this swarm technique, data points act like a flock of birds seeking food. Each agent remembers its Personal Best (pbest) discovery while communicating to track the flock's Global Best (gbest). Mathematical

	<i>equations adjust their "flight speed" (velocity) and position to pull them toward these optimal points.</i> 4. Real-World Applications: <i>These fast, flexible algorithms do not require perfectly structured data and are used to power complex logistics, routing, and hyper-parameter tuning in Deep Learning.</i>
Session 2	
<i>Topic</i>	<i>Introduction to Optimization Technique</i>
<i>Guest Name</i>	<i>Dr. Ravi Kagaju</i>
<i>Time</i>	<i>11:30-13:00</i>
<i>Topic Covered</i>	1. Foundations of Optimization: <i>Real-world applications and classical approaches.</i> 2. Core Solving Methods: <i>Graphical, Simplex, Two-Phase, and Big M Methods.</i>

	3. Understanding Solution Types: Basic, Basic Feasible, Optimal Basic Feasible, and Unbounded solutions.
Summary	1. Foundations of Optimization: Optimization uses strict mathematical formulas to maximize efficiency, such as minimizing costs or maximizing profits, and quietly powers major industries like airline scheduling and supply chains. 2. Core Solving Methods: Different problems require specific mathematical tools. The Graphical Method is highly visual but only works for problems with exactly two variables by plotting constraints on an X-Y graph. The Simplex Method is the standard for multi-variable linear problems, using matrix math to navigate multi-dimensional shapes to find the best corner. For problems where standard Simplex cannot find a starting point, Two-Phase and Big M Methods use heavily penalized artificial variables to force a valid beginning. 3. Understanding Solution Types: The

	algorithm evaluates different endpoints. Basic Solutions are mathematical intersections that might be physically impossible in the real world. Basic Feasible Solutions obey all real-world constraints but may not be the most efficient. The Optimal Basic Feasible Solution is the ultimate goal, providing the maximum return. An Unbounded Solution indicates a missing real-world constraint in the math, resulting in an impossible "infinite profit" scenario.
Session 3	
<i>Topic</i>	<i>India's E-Governance Journey and Fundamentals of Artificial Intelligence</i>
<i>Guest Name</i>	<i>Dr. V. S. Samy</i>
<i>Time</i>	<i>14:00-15:30</i>
<i>Topic Covered</i>	1. Building Blocks of Computation: Data Structures, Algorithms, and their relationship. 2. Artificial Intelligence Revolution: Defining intelligent systems and the AI/ML/DL evolution.

	3. Technology in Action: India's digital transformation and E-Governance.
Summary	1. The Backbone of Digital Systems: Dr. Sarma connected foundational computer science with intelligent computing, emphasizing that Data Structures and Algorithms (DSA) are not merely academic concepts. They serve as the critical engine powering both modern software and large-scale digital governance platforms. 2. Transforming Governance: By examining India's e-governance progress, the session demonstrated that digital transformation offers more than simple convenience. Moving government services online increases public transparency, cuts through bureaucratic red tape, and significantly reduces corruption.
Session 4	
Topic	Digital Transformation and Time Series Analysis using AI

<i>Guest Name</i>	<i>Dr. V. S Samay</i>
<i>Time</i>	<i>16:00-17:30</i>
<i>Topic Covered</i>	1. The Digital Evolution: Digitization, Digitalisation, and Digital Transformation. 2. Uncovering Patterns: Time-series basics, Wavelet Spectrum Analysis, and Behavioural Analysis. 3. Deep Learning for Sequence Data: RNN and LSTM networks.

Summary

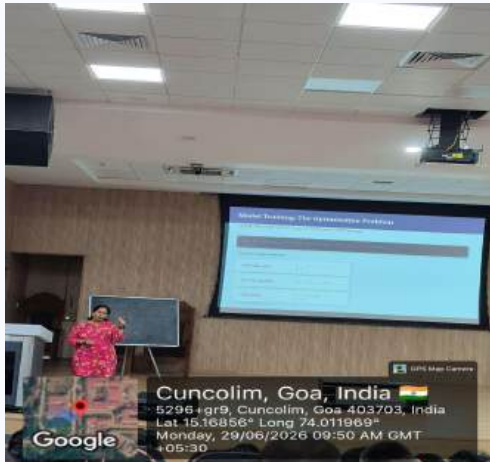
1. The Digital Evolution: Technological maturity progresses in clear stages. Digitization converts analog data to digital; Digitalisation uses that data to streamline workflows; and Digital Transformation is a fundamental, organization-wide shift driven entirely by digital tech.

2. Uncovering Patterns in Data: This transformation creates massive time-based datasets. Analysts break this down into long-term trends, seasonal predictability, and diurnal fluctuations. Wavelet Spectrum Analysis is an advanced mathematical tool used to identify the timing and frequency of unpredictable signals, while Behavioural Analysis studies user habits to predict future actions.

3. Deep Learning for Sequence Data: Standard analysis is sometimes insufficient, requiring specialized neural networks. Recurrent Neural Networks (RNN) possess short-term memory and process data sequentially, making them useful for basic text reading or forecasting. Long Short-Term Memory (LSTM) networks upgrade RNNs by using complex internal gates to decide exactly what information to retain long-term and what to forget, solving the "short-term

	<i>memory loss" issue.</i>
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<i>Day- Monday</i>	<i>Date- 29th June 2026</i>
<i>Session 1</i>	
<i>Topic</i>	<i>Model Training & Evaluation</i>
<i>Guest Name</i>	<i>Ms. Siddhi Naik</i>
<i>Time</i>	<i>09:30-11:00</i>



Topic Covered

The model lifecycle (training, validation, evaluation), fundamental regression (linear and logistic), geometric classifiers like Decision Trees and SVMs (including kernels and margins), and K-Nearest Neighbors (KNN).

Summary

1. The Model Lifecycle: Algorithms require a strict three-step process: training to learn hidden patterns, validation to fine-tune settings without exposing the test data, and evaluation on completely unseen data to ensure actual learning occurred rather than mere memorization.

2. Foundational Regression: Linear regression estimates continuous outcomes, while logistic regression determines probabilities for categorical, binary results.

3. Geometric and Proximity Classifiers: Decision Trees utilize transparent, flowchart-style rules to slice data. KNN bypasses complex math to classify data based purely on physical proximity to existing data points.

4. Advanced Mechanics: Support

	<i>Vector Machines (SVMs) seek the widest possible safety gap (margin) between data classes. They use kernel functions as a mathematical shortcut to map tangled data into higher dimensions for easier separation. Multiclass classification strategies upgrade these systems to sort three or more distinct categories.</i>
Session 2	
<i>Topic</i>	<i>Basic Structure of ANN</i>
<i>Guest Name</i>	<i>Mr. Devandra Kuvelkar</i>
<i>Time</i>	<i>11:30-13:00</i>
<i>Topic Covered</i>	<i>Real-world AI applications, Artificial Neural Network (ANN) architecture, perceptron geometry, and various activation functions including ReLU, Tanh, Softmax, and Sigmoid.</i>

Summary

1. Real-World Neural Networks:

ANNs drive modern AI, powering Natural Language Processing (NLP) chatbots, self-driving computer vision, and real-time financial fraud detection.

2. Architecture and Geometry:

The perceptron serves as the fundamental building block of a neural network. Geometrically, it acts as a linear classifier attempting to draw a perfectly straight line or flat plane to separate data categories.

3. Activation Functions:

These mathematical triggers introduce vital non-linearity, allowing networks to learn complex, curved patterns. Sigmoid scales outputs between 0 and 1 but can cause a "vanishing gradient" slowdown in deep networks. Tanh centers outputs at zero, making training faster and more stable than Sigmoid. ReLU (Rectified Linear Unit) is the computationally efficient industry standard for hidden layers, outputting zero for negative inputs and the raw number for positive ones. Softmax is reserved for final output layers, smoothly scaling raw numbers into a

	<i>list of probabilities that total 100%.</i>
Session 3	
<i>Topic</i>	<i>Hands on Linear Regression</i>
<i>Guest Name</i>	<i>Ms Siddhi Naik</i>
<i>Time</i>	<i>14:00-15:30</i>
<i>Topic Covered</i>	<i>Classification metrics (Confusion Matrix, Recall, Precision, Accuracy, F1-Score, ROC Curve), supervised classifiers (KNN, Logistic Regression), and the mathematics of Decision Tree splits (Entropy, Gini Impurity, Information Gain).</i>

Summary

1. Classification Evaluation: Relying solely on accuracy is dangerous; a complete picture requires multiple metrics. The Confusion Matrix maps exact true/false positives and negatives. Precision checks accuracy when predicting "yes," while Recall checks how many actual "yeses" were successfully identified from the dataset. The F1-Score balances both into a single metric, making it vital for imbalanced datasets. The ROC Curve visually plots class separation to help locate the ideal threshold.

2. Core Classifiers: A practical recap of KNN's proximity-based classification and Logistic Regression's mathematical boundaries.

3. Decision Tree Splits: Trees determine branching using Entropy and Gini Impurity, which measure data randomness or "messiness". The goal is to achieve the lowest possible impurity. Information Gain dictates the actual split by calculating how much "cleaner" the data becomes, subtracting the weighted entropy of the new branches from the original data's

	<i>messiness.</i>
Session 4	
<i>Topic</i>	<i>Hands on Evaluation Metrics in ML & Simple Linear Regression</i>
<i>Guest Name</i>	<i>Mr. Devendra Kuvelkar</i>
<i>Time</i>	<i>16:00-17:30</i>
<i>Topic Covered</i>	<i>Practical implementation of simple linear regression, applying evaluation metrics using Python, and generating confusion matrices and F1 scores.</i>
<i>Summary</i>	1. Simple Linear Regression: <i>Transitioning from theory to active coding, students trained foundational models to predict continuous outcomes by mathematically fitting the best possible straight line through data points.</i> 2. Hands-On Metrics: <i>Once a model is trained, it must be proven effective on unseen data. Participants utilized</i>

Python libraries, such as Scikit-Learn, to programmatically calculate hard accuracy scores and error rates.

3. Deep Dive into Metrics: *Writing code to generate a Confusion Matrix is crucial because it maps out exactly how a model failed by detailing false positives versus false negatives. Implementing the F1 Score is the ultimate tie-breaker for imbalanced data, calculating a harmonic mean between Recall and Precision to provide a highly realistic grade of a model's true predictive power.*

Day- Tuesday	Date- 30th June 2026
Session 1	
Topic	Introduction to Cryptography
Guest Name	Ms Siddhi Naik
Time	09:30-11:00
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<i>Topic Covered</i>	<i>The fundamentals of cryptography and the necessity of information security. It covered key security services including authentication, access control/authorization, data confidentiality, data integrity, non-repudiation, and availability. Various cryptographic techniques were explored, with a focus on public-key cryptography.</i>
<i>Summary</i>	1. The Security Landscape: <i>In a digital world where data is a primary asset, information security is vital to protect against unauthorized access and corruption. Cryptography serves as the mathematical foundation, scrambling readable plaintext into unreadable ciphertext so that only authorized keyholders can access it.</i> 2. Six Pillars of Security Services: <i>A secure system relies on verifying identity (Authentication), controlling permissions (Authorization), hiding secrets from interceptors (Confidentiality), preventing data tampering (Integrity), proving digital actions (Non-repudiation), and ensuring constant system access for users (Availability).</i> 3. Cryptographic Techniques: <i>Traditional (Symmetric) Cryptography utilizes a single,</i>

	<i>shared key for both locking and unlocking, which is fast but risky to share. Public Key (Asymmetric) Cryptography secures the modern internet by utilizing a matched pair: a widely shared Public Key to lock data and a completely secret Private Key to unlock it.</i>
Session 2	
<i>Topic</i>	<i>Artificial Neural Networks</i>
<i>Guest Name</i>	<i>Mr. Devandra Kuvelkar</i>
<i>Time</i>	<i>11:30-13:00</i>
<i>Topic Covered</i>	<i>Neural Network architecture mechanics, including input layer dimensionality and normalization. The session covered the loss function, gradient descent optimization (updating weights and learning rates), forward and backward passes, the chain rule, and visualizing model underfitting versus overfitting.</i>

Summary

1. Data Input: The network's input layer is defined by its dimensionality (number of incoming features). Normalization is applied to scale this data down, ensuring larger numbers do not disrupt the mathematical processing.

2. The Engine of Learning: During a Forward Pass, data moves through the network to generate a prediction. The Loss Function calculates the exact error of that prediction, and a Backward Pass (backpropagation) sends this error signal back to adjust internal weights.

3. Optimization: Gradient Descent continuously calculates the error's slope and updates weights to minimize loss. The Chain Rule manages the calculus for distributing errors backward, while the Learning Rate dictates the size of the adjustment steps.

4. Model Diagnostics: It is crucial to visually evaluate if a model genuinely learned the

	<i>patterns (Optimal Fit), blindly memorized the training material (Overfitting), or failed to learn entirely (Underfitting).</i>
Session 3	
<i>Topic</i>	<i>Hands on Regression</i>
<i>Guest Name</i>	<i>Mr. Devendra Kuvelkar</i>
<i>Time</i>	<i>14:00-15:30</i>
<i>Topic Covered</i>	<i>Practical applications of Multiple Regression, Logistic Regression, and Naive Bayes models in real-world scenarios.</i>

Summary

1. Multiple Regression: This model is highly effective in asset management for tasks requiring the simultaneous evaluation of multiple continuous financial data streams, such as forecasting tax liabilities or calculating real-time depreciation.

2. Logistic Regression: Serving as the backbone for automated compliance, this algorithm excels at binary probabilities, allowing systems to instantly flag audits or verify if a corporate entity meets specific economic presence criteria.

3. Naive Bayes: In document parsing, this algorithm functions as an ultra-fast text sorter. It rapidly categorizes unstructured incoming documents to keep databases organized.

4. The Preprocessing Pipeline: Real-world data is inherently messy. Before deploying predictive algorithms in production environments, the data must undergo rigorous cleaning, normalization, and

	<i>vectorization to prevent system failures.</i>
Session 4	
<i>Topic</i>	<i>Hands on Classification Techniques</i>
<i>Guest Name</i>	<i>Ms. Siddhi Naik</i>
<i>Time</i>	<i>16:00-17:30</i>
<i>Topic Covered</i>	<i>A practical comparison and implementation of supervised learning classifiers, focusing on Decision Trees (DT), Support Vector Machines (SVM), and K-Nearest Neighbors (KNN).</i>

Summary

1. Decision Trees (DT): Operating via a flowchart-like structure of rules, this model is highly interpretable and ideal for generating blocky, grid-like boundaries when stakeholders require clear explanations for predictions.

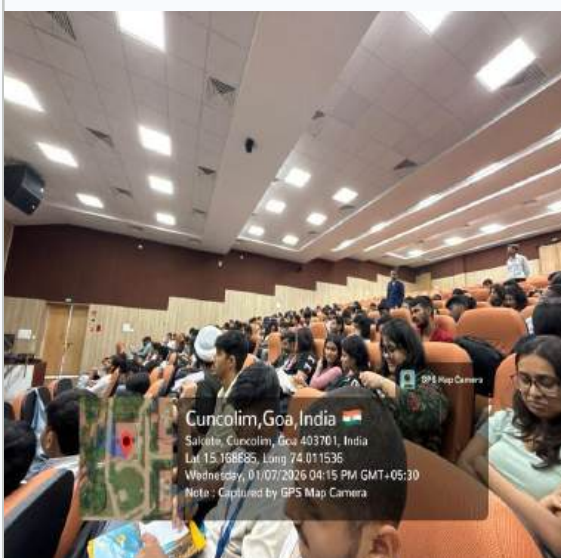
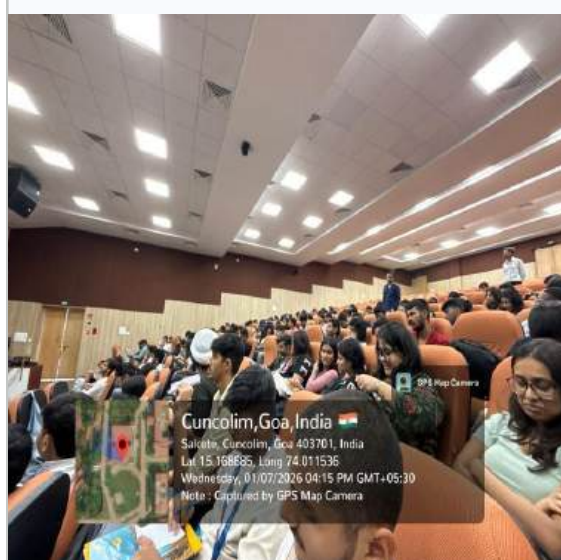
2. Support Vector Machines (SVM): Utilizing mathematical tricks to manage complex datasets, SVM searches for the optimal geometric boundary (margin) to separate classes. It excels at drawing smooth, precise lines and performs well even when features outnumber data points.

3. K-Nearest Neighbors (KNN): This simple yet effective proximity-based voting system creates localized, island-like boundaries by classifying new data strictly based on the identities of its closest neighbors.

4. Model Selection: No single model is universally superior. DTs provide transparency, SVMs navigate complex and

	<i>overlapping data, and KNN serves as a rapid, assumption-free baseline model.</i>
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<i>Day- Wednesday</i>	<i>Date- 1st July 2026</i>
<i>Session 1</i>	
<i>Topic</i>	<i>Sensors & their Engineering Application</i>
<i>Guest Name</i>	<i>Prof. Saidi Reddy</i>
<i>Time</i>	<i>09:30-11:00</i>



Topic Covered

Definitions of sensors, transducers, and actuators within the AI/ML/DL space, expanding market caps, and sensor applications (smart, active, passive). It also covered characteristic curves, the differences between accuracy, precision, and resolution, and fiber versus copper

	<i>transmission media.</i>
Summary	1. Sensor Performance Metrics: <i>Understanding a sensor's capabilities requires distinguishing between Accuracy (how close the reading is to the true value), Precision (the consistency of repeated measurements), and Resolution (the smallest detectable physical change).</i> 2. The Characteristic Curve: <i>This calibration graph plots physical input against electrical output. It reveals crucial behaviors such as Linearity, Sensitivity (output change per unit of input), and Hysteresis (reading discrepancies depending on whether values are rising or falling).</i> 3. Transmission Media: <i>Once data is captured, it must be transmitted. Copper cables are inexpensive and can carry power, but they suffer from distance degradation and Electromagnetic Interference (EMI). Fiber Optic cables transmit via light pulses, offering massive bandwidth, near-zero distance loss, and</i>

	<i>complete EMI immunity.</i>
Session 2	
<i>Topic</i>	<i>CNN</i>
<i>Guest Name</i>	<i>Mr. Devandra Kuvelkar</i>
<i>Time</i>	<i>11:30-13:00</i>
<i>Topic Covered</i>	<i>Introduction to Convolutional Neural Networks (CNN), basic working examples, and core architecture components including Convolutional, Pooling, Flattening, and Fully Connected layers. The session also detailed the ReLU activation function, ANN integration, and key properties like local connectivity, parameter sharing, hierarchical feature learning, and translation invariance.</i>

Summary	1. The Architecture of a CNN: The Convolutional Layer slides mathematical filters over an image to detect features like edges and textures. The Pooling Layer then compresses this feature map to reduce computational load while retaining essential data. After applying the ReLU Activation Function to introduce non-linearity, the 2D spatial features are transformed into a 1D list via Flattening. This list is fed into a traditional Fully Connected network to make final classifications. 2. Key Principles & Properties: CNNs utilize Local Connectivity (neurons only examine localized image patches) and Parameter Sharing (the same filter weights are used across the entire image to drastically reduce required calculations). Translation Invariance allows the network to recognize objects regardless of their position in the frame, while Hierarchical Feature Learning enables the model to build complex object recognition in stages, progressing from basic lines to complete shapes.
Session 3	
Topic	Cyber Security

<i>Guest Name</i>	<i>Ms. Siddhi Naik</i>
<i>Time</i>	<i>14:00-15:30</i>
<i>Topic Covered</i>	<i>Introduction to Cyber Security, the CIA Triad, and an overview of malware and network attacks (Phishing, DoS, MITM, Trapdoors, Logic Bombs, Trojans, Zombies, Viruses, Ransomware, SQL Injection). The session also covered defensive architectures like Antivirus, Firewalls, Vulnerability Scanners, and SIEM, concluding with the offensive and defensive uses of AI.</i>
<i>Summary</i>	1. Fundamentals & The CIA Triad: Cyber Security aims to protect digital ecosystems from exploitation and disruption. It is built upon the CIA Triad: Confidentiality (keeping data hidden from unauthorized users), Integrity (ensuring data remains accurate and untampered), and Availability (guaranteeing legitimate users can access systems). 2. Cyber Threats & Malware Anatomy: Systems are compromised through varied vectors. Network attacks include Phishing, DoS/DDoS, and SQL Injections, while malware exploits include self-replicating Viruses, disguised Trojan Horses, extortion-based Ransomware,

	<i>remote-controlled Zombies, and condition-triggered Logic Bombs.</i> 3. Defensive Architecture & AI: <i>Organizations use multi-layered defenses, including Firewalls, signature-based Antivirus software, and Security Information & Event Management (SIEM) engines that aggregate network-wide logs to spot anomalies. AI is increasingly deployed Defensively for high-speed automated threat hunting, while attackers utilize Offensive AI to generate dynamically changing malware and scalable phishing campaigns.</i>
Session 4	
<i>Topic</i>	<i>Hands-on Convolutional Neural Networks (CNN) and Artificial Neural Networks (ANNs)</i>
<i>Guest Name</i>	<i>Mr. Devendra Kuvelkar</i>
<i>Time</i>	<i>16:00-17:30</i>

<i>Topic Covered</i>	<i>CNN factors, problem perception, CNN workflow examples, evaluating Epochs vs. Accuracy and Epochs vs. Loss graphs, Python coding practice for CNN numericals, and a comparative summary of CNN and ANN topics.</i>
<i>Summary</i>	1. Problem Formulation & CNN Workings: <i>The session focused on perceiving real-world visual problems and structuring data appropriately for deep learning. It included a complete walkthrough of how convolutional, pooling, and fully connected layers interact to extract features and generate predictions.</i> 2. Performance Evaluation Graphs: <i>Model learning was monitored over training cycles (Epochs). The Epochs vs. Accuracy graph tracks prediction improvement, while the Epochs vs. Loss graph tracks mathematical error, serving as a critical diagnostic tool to identify overfitting or bad learning rates.</i> 3. Hands-On Python Implementation: <i>Students translated theoretical CNN numericals into working Python code,</i>

allowing them to practically build, compile, and train actual neural network models.

4. Comprehensive Summary: *A concluding comparison reinforced the core distinction: ANNs serve as generalized networks for tabular data, while CNNs utilize localized filters and weight sharing specifically optimized for visual and spatial tasks.*

Day- Thursday	Date- 2nd July 2026
Session 1	
Topic	<i>Reinforcement Learning Methods applied to Job-shop scheduling programming</i>
Guest Name	<i>Dr. Lokesh Bramhane</i>
Time	<i>09:30-11:00</i>
	
	

<i>Topic Covered</i>	<i>A brief overview of Machine Learning types, Reinforcement Learning (RL), the Markov Decision Process (MDP), Active versus Passive RL, and the application of RL in various domains.</i>
<i>Summary</i>	1. Types of Machine Learning (Brief): A quick recap of ML paradigms, highlighting Supervised Learning (learning from labeled examples), Unsupervised Learning (finding structures in raw data), and Reinforcement Learning (interacting with an environment to maximize rewards). 2. The Core of RL: The mathematical foundation of RL is the Markov Decision Process (MDP). An MDP involves States (the environment's current situation), Actions (choices the agent can make), and Rewards (feedback received). The agent's objective is to maximize long-term rewards. 3. Active vs. Passive RL: In Passive RL, the agent evaluates a pre-given, fixed set of rules (policy) without making its own choices. In Active RL, the agent is placed in an environment without instructions and must learn the optimal policy entirely

	through trial and error. 4. RL in Action: The session applied RL to Job-Shop Scheduling, a complex manufacturing problem. By modeling the factory floor as an MDP, the RL agent can dynamically adjust schedules in real-time—such as routing around machine breakdowns—to minimize total production time efficiently, unlike traditional static algorithms.
Session 2	
<i>Topic</i>	<i>Introduction to AI and GEN AI</i>
<i>Guest Name</i>	<i>Dr. Parvati Swain</i>
<i>Time</i>	<i>11:30-13:00</i>
<i>Topic Covered</i>	<i>Fairness, bias, and Human-in-the-Loop (HITL) principles. The history of AI and vision networks including AlexNet, ZFNet, VGGNet, GoogLeNet, and ResNet. Introductions to Generative AI, Large Language Models (LLMs), GANs, and Diffusion Models.</i>

Summary

1. Ethical Foundations: The session emphasized that AI models scale the systemic biases present in their historical training data. To mitigate this, a Human-in-the-Loop (HITL) design pattern is critical, where human reviewers continuously monitor and validate model outputs to catch hallucinations and ensure compliance.

2. The Evolution of Vision: Before GenAI, deep learning focused on deep vision architectures. AlexNet and ZFNet proved the dominance of convolutional networks; VGGNet utilized deep, simple filters; and GoogLeNet introduced parallel processing. Ultimately, ResNet solved the "vanishing gradient" mathematical bottleneck using "skip connections," allowing networks to scale to hundreds of layers.

3. The Generative Frontier: The focus shifted to GenAI and LLMs, which generate entirely novel text and code using massive Transformer models. The session also covered GANs (a two-model system where a generator and discriminator compete to create synthetic data) and Diffusion Models (the math behind modern image generation

	<i>that works by systematically erasing noise).</i>
Session 3	
<i>Topic</i>	<i>Cultural Event</i>
<i>Guest Name</i>	(N/A)
<i>Time</i>	14:00-17:30
<i>Topic Covered</i>	<i>Cultural Event (Covers both Session 3 and Session 4 time slots).</i>
<i>Summary</i>	<i>Participants engaged in a Cultural Event for the remainder of the afternoon program.</i>
Session 4	
<i>Topic</i>	<i>(Combined with Session 3 Cultural Event)</i>
<i>Guest Name</i>	(N/A)
<i>Time</i>	(N/A)

<i>Topic Covered</i>	(N/A)
<i>Summary</i>	(N/A)

Day- Friday	Date- 3rd July 2026
Session 1	
Topic	Recap on Machine Learning
Guest Name	Dr. Tarun Jaiswal
Time	09:30-11:00
	
	

<i>Topic Covered</i>	<i>An overview of Machine Learning fundamentals, covering its three primary pillars (Supervised, Unsupervised, and Reinforcement Learning). The session also detailed data optimization techniques like Feature Engineering and Dimensionality Reduction, the Generalization Gap (underfitting vs. overfitting), and the necessity of Explainable AI (XAI) to solve the Black Box problem.</i>
<i>Summary</i>	1. Core Concepts & The Three Pillars: <i>Machine Learning involves algorithms identifying patterns directly from data rather than relying on explicit programming. The field is built on three pillars: Supervised Learning (training on labeled examples), Unsupervised Learning (discovering structures in unlabeled data), and Reinforcement Learning (learning via trial and error for maximum reward).</i> 2. Data Preparation & Optimization: <i>Data must be engineered prior to model training. Feature Engineering involves manually selecting or creating relevant data inputs, while Dimensionality Reduction compresses large datasets by removing the least important variables to save computing power without losing meaning.</i>

	3. The Generalization Gap: A model's success is determined by its performance on unseen data. The goal is to minimize the "Generalization Gap," which is the difference in accuracy between training performance and real-world application, avoiding both underfitting (too simplistic) and overfitting (perfect memorization of training data). 4. Trust in AI (XAI): Advanced models often suffer from the "Black Box Problem," where their internal math is too complex for humans to decipher. Explainable AI (XAI) provides frameworks that force models to generate human-readable evidence for their decisions, a critical requirement for compliance in industries like healthcare and finance.
Session 2	
<i>Topic</i>	<i>Generative AI: Future Directions & Emerging Applications</i>

<i>Guest Name</i>	<i>Prof. Sriparna Saha</i>
<i>Time</i>	<i>11:30-13:00</i>
<i>Topic Covered</i>	<i>The evolution of Generative AI and Large Language Models (LLMs), advanced Prompt Engineering techniques (Zero-shot, Few-shot, CoT, ToT, PoT), and GenAI applications like text summarization and PDF analysis. The session also explored Multimodal AI, Blockchain intersections, and Responsible AI usage.</i>
<i>Summary</i>	1. GenAI & LLM Evolution: Generative AI shifts the focus from classifying data to creating entirely new content, driven by Large Language Models (LLMs) built on the context-heavy Transformer architecture. 2. Advanced Prompt Engineering: Prompts function as code to direct LLMs. Users can set strict templates using Zero-shot or Few-shot techniques, or apply structural reasoning frameworks like Chain-of-Thought (CoT) for step-by-step logic, Tree-of-Thoughts (ToT) for evaluating multiple paths, and Program-of-Thoughts (PoT) to offload

	<i>math processing to Python scripts.</i> 3. Multi-Domain AI Tools: GenAI is automating document analysis and accelerating research through platforms like Semantic Scholar (mapping academic literature) and Vizly (turning text queries into live data visualizations). 4. Multimodal, Blockchain, & Ethics: Future systems are Multimodal, combining mixed inputs like text-to-image generation. The technology is also intersecting with Blockchain to allow autonomous AI agents to execute decentralized transactions. Finally, deploying Responsible AI requires strict guardrails to prevent data bias and factual hallucinations.
Session 3	
<i>Topic</i>	<i>Introduction to Blockchain Technology</i>
<i>Guest Name</i>	<i>Ms. Siddhi Naik</i>

<i>Time</i>	14:00-15:30
<i>PHOTO 1: Speaker (Geo Tag)</i>	<i>PHOTO 2: Speaker (Geo Tag)</i>
<i>PHOTO 3: Audience (Geo Tag)</i>	<i>PHOTO 4: Audience (Geo Tag)</i>
<i>Topic Covered</i>	<i>Comparisons between traditional banking and blockchain. The session covered foundational concepts (ledgers, transparency, immutability), components (Blocks, Nodes, Hashes, Miners), types of blockchains, consensus mechanisms, smart contracts, and real-world applications.</i>
<i>Summary</i>	1. Blockchain Foundations & Architecture: Blockchain functions as a decentralized, immutable digital ledger distributed across a peer-to-peer network of independent computers (nodes), unlike centralized traditional banking. Its core characteristics ensure Transparency, Immutability (tamper-proofing), and Decentralization. Network types range from open Public chains (like Bitcoin) to restricted Private and Consortium chains. 2. The Mechanics of Blockchain: Data is organized into Blocks, each containing a unique cryptographic Hash of the

	<i>preceding block to create an unbreakable Chain. Merkle Trees compress and verify the block's transaction data. The standard operational flow involves a transaction request being broadcast, validated by nodes, bundled into a block via mining (solving cryptographic puzzles like SHA-256), and permanently appended to the ledger.</i>
Session 4	
<i>Topic</i>	<i>Object Detection Using Deep Learning YOLO v5 Model</i>
<i>Guest Name</i>	<i>Mr. Devendra Kuvelkar</i>
<i>Time</i>	<i>16:00-17:30</i>
<i>Topic Covered</i>	<i>Computer vision using the YOLO (You Only Look Once) model architecture (Backbone, Neck, Head), the Darknet framework, historical CNN architectures, and Transfer Learning. The second half covered Brain-Computer Interfaces (BCI), signal acquisition methods (EEG, ECOG), feature extraction, and ML's role in translating brainwaves.</i>

Summary

Part 1: Computer Vision: YOLO enables real-time object detection via a single neural network pass, using a Backbone to extract features, a Neck to mix them, and a Head to predict bounding boxes, all trained on the optimized Darknet framework. The session reviewed the evolution of CNNs from LeNet-5 to ResNet, emphasizing Transfer Learning, which saves compute power by fine-tuning heavy, pre-trained models for new tasks.

Part 2: Brain-Computer Interfaces (BCI): BCIs connect the brain to external machines via a strict pipeline: Signal Acquisition, Preprocessing, Feature Extraction, Classification, and Device Command. Signals are acquired through methods ranging from non-invasive EEG scalp sensors to highly accurate, invasive ECOG electrodes. Machine Learning acts as the translator, finding patterns in the extracted frequency and amplitude features to classify user intent (like imagining limb movement), which powers applications such as robotic prosthetics and neuro-rehabilitation.

Day- Saturday	Date- 4th July 2026
Session 1	
Topic	Feedforward Neural Network
Guest Name	Dr. Tarun Jaiswal
Time	09:30-11:00
 	
 	

<i>Topic Covered</i>	<i>Fundamentals of Feedforward Neural Networks (FNNs) and biological inspiration. Core components including the Perceptron, weights, bias, summation, and non-linear activation functions (Sigmoid, ReLU, Softmax). The session also covered multi-layer architecture, forward propagation, loss functions, gradient descent, backpropagation, and the Universal Approximation Theorem.</i>
<i>Summary</i>	1. Core Mechanics: FNNs are foundational, biologically-inspired deep learning models where data strictly moves forward from input to output. The base unit is the Perceptron, which calculates a weighted sum of inputs and adds a bias to provide flexibility, represented mathematically as $z = \sum(w_i x_i) + b$. 2. Activation & Architecture: This summed total is passed through an activation function, such as $a = f(z)$, to introduce crucial non-linearity; otherwise, the network could not learn complex patterns. Common functions include ReLU for computational efficiency in hidden layers and Softmax for generating probabilities in output layers. These perceptrons are organized into Input, Hidden, and Output layers.

	3. Training & Evaluation: The network learns by calculating the error of its predictions using a Loss Function. It minimizes this error through Gradient Descent, which adjusts weights step-by-step at a specific Learning Rate. This process is executed via Backpropagation, which calculates errors backward through the network. While the Universal Approximation Theorem states these networks can approximate any continuous function, engineers must carefully balance model complexity to avoid underfitting (failing to learn) or overfitting (memorizing data but failing on new inputs).
Session 2	
<i>Topic</i>	<i>Introduction to CNN</i>
<i>Guest Name</i>	<i>Dr. Tarun Jaiswal</i>
<i>Time</i>	<i>11:30-13:00</i>
<i>Topic Covered</i>	<i>The Flat Vector Problem and how Convolutional Neural Networks (CNNs) solve it. Detailed exploration of Convolutional Layers (filters, feature maps), Pooling Layers (down-sampling), dimension controls (stride, zero padding), and final processing</i>

	(flattening, dense networks, Softmax).
Summary	1. The Flat Vector Problem: Standard neural networks struggle with images because flattening a 2D image into a 1D array destroys vital spatial context and leads to weight explosion. Inspired by the human visual cortex, CNNs solve this by automating feature extraction and maintaining distortion invariance with significantly fewer parameters. 2. Core Architecture: The Convolutional Layer utilizes mathematical filters that sweep across the image to generate a Feature Map. Engineers control this process using Stride Length (step size) and Zero Padding (preserving edge data). Next, the Pooling Layer down-samples the feature map to consolidate the most important visual data and reduce size. 3. Final Processing: After cycling through convolution and pooling, the final 2D maps undergo Flattening into a 1D array. This array is fed into standard Fully Connected layers for Dense Processing, ultimately using a Softmax output to provide probability-based classifications. This

	<i>architecture serves as the foundation for complex tasks like Semantic Segmentation and Object Detection.</i>
Session 3	
<i>Topic</i>	<i>Autoencoders</i>
<i>Guest Name</i>	<i>Dr. Tarun Jaiswal</i>
<i>Time</i>	<i>14:00-15:30</i>
<i>Topic Covered</i>	<i>The purpose and architecture of Autoencoders, including encoders, decoders, and latent space. The session explored Denoising Autoencoders, core applications like dimensionality reduction and feature variation, and advanced constraints like sparse activation.</i>

Summary

1. Introduction & Purpose: An Autoencoder is an unsupervised network built for efficient data compression. It features an encoder that compresses input data into a smaller "bottleneck" (latent space), and a decoder that reconstructs the original input from that compressed state. This allows the automatic extraction of meaningful information from massive, unlabeled datasets.

2. Data Robustness: Real-world data is inherently noisy. To prevent the model from simply memorizing flaws, engineers use Denoising Autoencoders. By intentionally feeding the network corrupted data and calculating loss against clean data, the network is forced to learn robust, underlying patterns rather than copying noise.

3. Applications & Architecture: Autoencoders are heavily used for non-linear dimensionality reduction (compressing massive datasets) and feature variation (morphing compressed data to generate new outputs). To ensure the network specializes rather than just memorizing data flows, engineers apply Sparse Activation, a mathematical penalty that allows only a small percentage of hidden neurons to fire, forcing

	<i>each to learn highly distinct features.</i>
Session 4	
<i>Topic</i>	<i>IOT</i>
<i>Guest Name</i>	<i>Dr. Lokesh Bramhane</i>
<i>Time</i>	<i>16:00-17:30</i>
<i>Topic Covered</i>	<i>Introduction to the Internet of Things (IoT), sensors, and actuators. A deep dive into Integrated Circuit (IC) testing, covering the four possible outcomes (True Pass, True Reject, Test Escape, Yield Loss), collaborative testing responsibilities, and implicit versus explicit testing methods.</i>

Summary

1. IoT and Hardware Foundations: IoT relies on digital systems interacting with the physical world via sensors and actuators. However, the foundational hardware—Integrated Circuits (ICs)—must function perfectly, making rigorous testing critical to modern manufacturing.

2. Testing Outcomes: When evaluating an IC, there are four results: True Pass (good device passes), True Reject (defective device fails), Test Escape (a critical failure where a bad device sneaks through), and Yield Loss (a wasteful error where a good device is rejected).

3. Collaborative Responsibility: Preventing Test Escapes and Yield Loss is a supply chain-wide responsibility. It requires Design and EDA teams to ensure the chip is testable, Fabrication facilities to manufacture it, and specialized Test and Reliability Engineers to run diagnostics. This involves both implicit (structural integrity) and explicit (functional behavior) testing, culminating in a high-volume Production Test on the factory floor to guarantee hardware quality.

Day- Sunday	Date- 5th July 2026
Session 1	
Topic	<i>The transformers architecture</i>
Guest Name	<i>Dr. Tarun Jaiswal</i>
Time	<i>09:30-11:00</i>
	



Topic Covered

The sequential bottlenecks of RNNs and the core Transformer architecture (Encoder/Decoder, Tokenization, Positional Encoding). Detailed breakdown of the Attention Mechanism (Self/Cross attention, QKV, Multihead), internal mechanics (Masked Self-Attention, Add & Norm), and final execution lifecycles.

Summary

1. Solving Sequential Bottlenecks: Transformers were developed to fix the limitations of older Recurrent Neural Networks (RNNs), which processed text word-by-word, leading to slow training and context loss over long paragraphs. Transformers process entire sentences simultaneously using an Encoder (to understand input) and a Decoder (to generate output). Text is first broken down via Tokenization and mapped into vectors using Input Embeddings, while Positional Encoding injects mathematical timestamps so the model retains word order.

2. The Attention Mechanism: The core engine is Attention, which allows words to focus on neighboring words for context (Self-Attention) or look back at the Encoder's output (Cross-Attention). This relies on the QKV (Query, Key, Value) mechanism—similar to searching a library catalog to retrieve specific book contents. Multihead Attention runs this process in parallel to simultaneously grasp grammar, tone, and context.

3. Internal Mechanics & Execution: The Decoder uses Masked Self-Attention as a "blindfold" to prevent the model from cheating by looking at future words during training. Add & Norm layers act as residual connections to prevent data loss. Finally, the data passes through a Feed Forward Neural Network, a Linear layer, and Softmax to

	<i>output human-readable probabilities. The model learns rapidly via parallel "teacher forcing" during training, but must generate text one word at a time (auto-regressively) during live inference.</i>
Session 2	
<i>Topic</i>	<i>Brain Computer Interface using ML & it's domain of research</i>
<i>Guest Name</i>	<i>Dr. Damodar Reddy</i>
<i>Time</i>	<i>11:30-13:00</i>
<i>Topic Covered</i>	<i>Introduction to Brain-Computer Interfaces (BCI), signal acquisition modalities (Invasive, ECOG, EEG), and the core pipeline (Preprocessing, Feature Extraction, Classification). Categorization of systems (Active, Reactive, Passive) and the transition to Deep Learning and Transfer Learning for applications like neurorehabilitation and prosthetics.</i>

Summary

1. BCI Fundamentals & Acquisition: A BCI provides a direct communication pathway from the human brain to external devices. Signal Acquisition is critical: Invasive BCIs use surgically implanted microelectrodes for zero interference but high risk; Semi-Invasive BCIs use Electrocorticography (ECOG) grids resting on the brain surface; and Non-Invasive BCIs use wearable EEG caps, which are safe but capture faint, noisy signals.

2. The ML Pipeline: Because brainwaves are noisy, the data requires rigorous Preprocessing (artifact removal). Machine Learning then performs Feature Extraction to isolate mathematical markers and Classification to decode the user's intent, triggering an Output Interaction. These systems can be Active (intention-driven), Reactive (stimulus-driven), or Passive (monitoring cognitive states). Modern research relies heavily on Deep Learning and Transfer Learning to overcome data scarcity.

3. Domains of Research: The primary focus is Medical & Neurorehabilitation. ML-powered BCIs decode motor imagery (the neural blueprints created when a paralyzed user imagines moving) to control robotic prosthetics intuitively. They also provide assistive communication for individuals with locked-in syndrome and are expanding into

	<i>non-medical sectors like hands-free gaming and pilot safety monitoring.</i>
Session 3	
<i>Topic</i>	<i>Evaluation Test</i>
<i>Guest Name</i>	<i>(N/A)</i>
<i>Time</i>	<i>14:00-15:30</i>
<i>Topic Covered</i>	<i>All topics covered till date.</i>
<i>Summary</i>	<i>Students completed a proctored, 50-mark examination that evaluated their comprehension of all topics and materials taught throughout the preceding sessions of the program.</i>
Session 4	
<i>Topic</i>	<i>Valedictory Session</i>
<i>Guest Name</i>	<i>(N/A)</i>
<i>Time</i>	<i>16:00-17:30</i>



Topic Covered

Valedictory Session and concluding remarks.

Summary

The program concluded with a brief address by the host to introduce the guests. Faculty members from NIT Goa formally felicitated the faculty coordinators from MIT WPU, and vice versa. The event finalized with several students sharing their personal experiences and takeaways from the immersion program.

